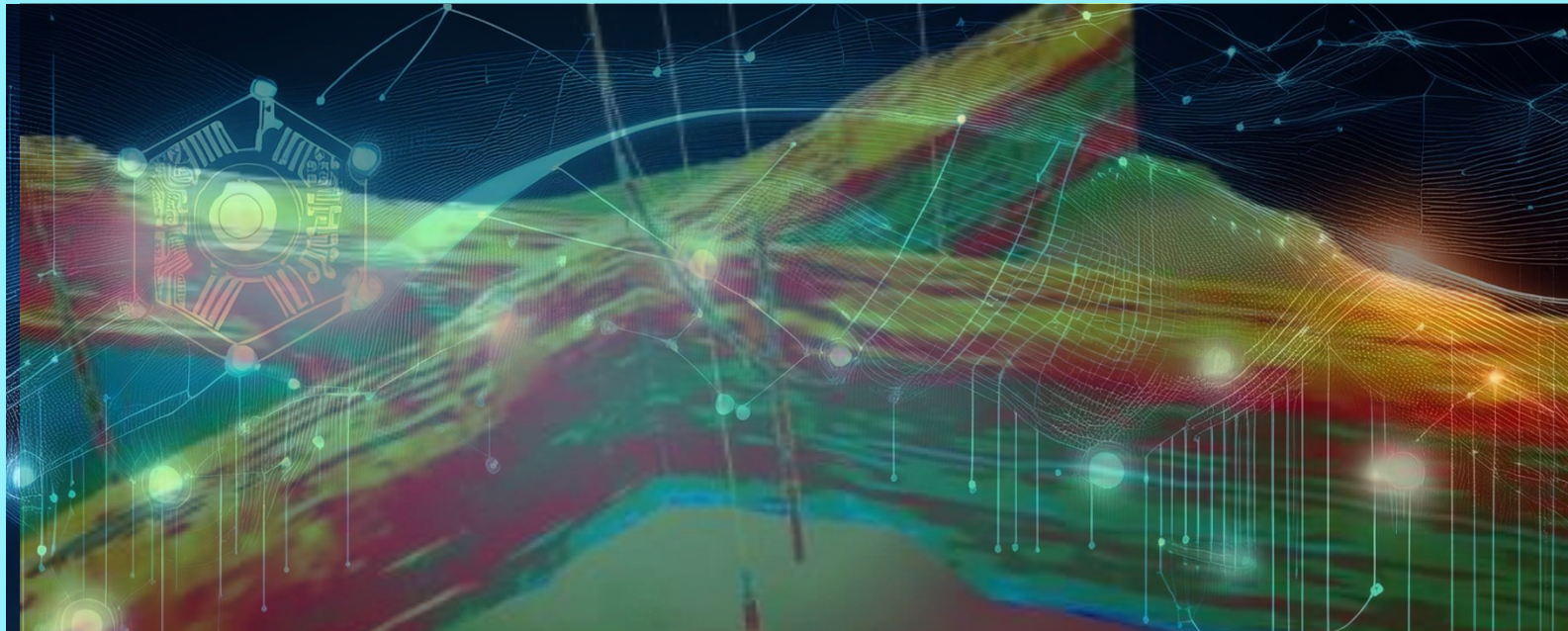
Machine Learning for Geoscientists

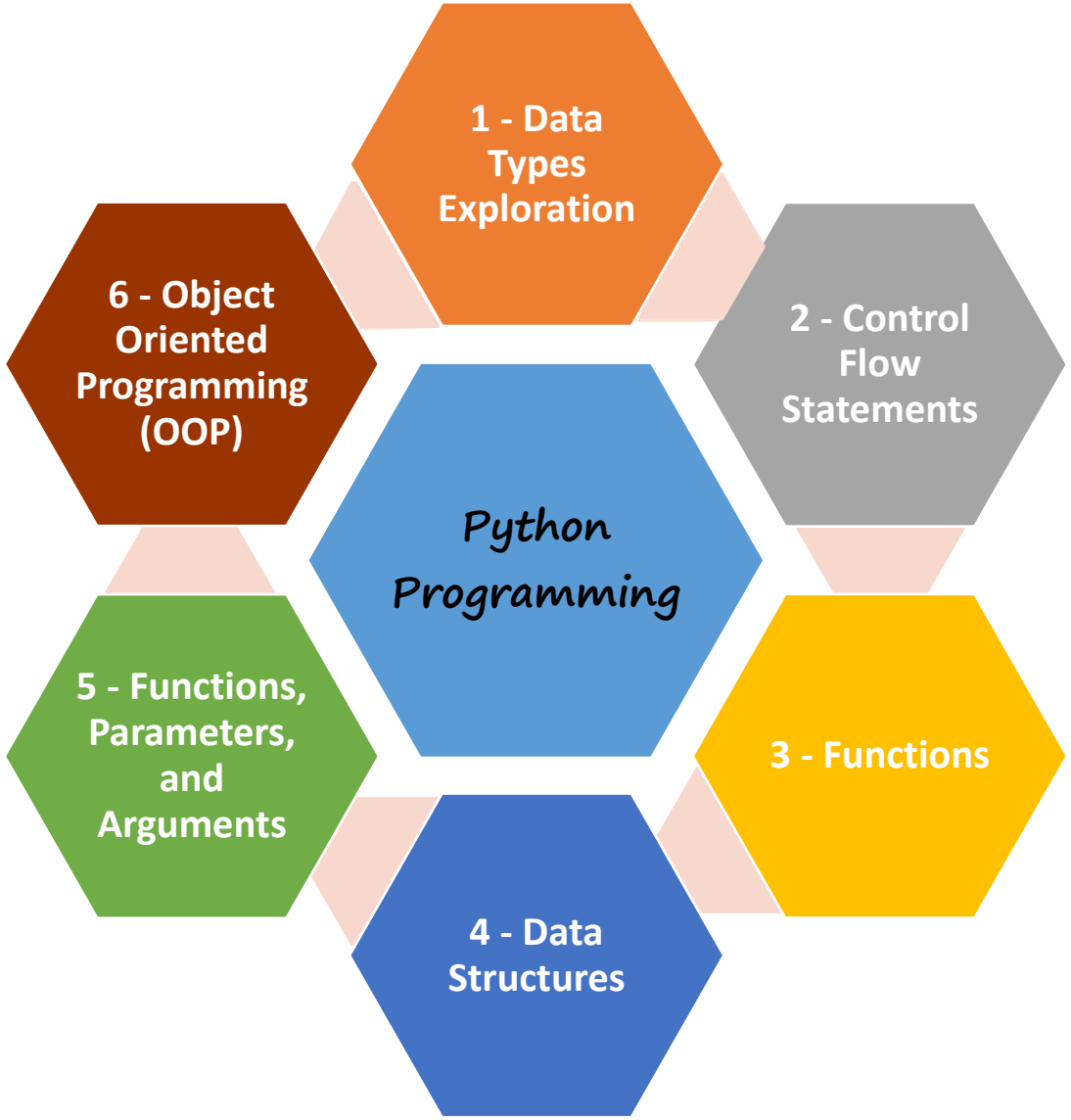
Theoretical Intuition and Hands-on Applications

Module 2: Python Programming

Ahmed Khairy
Senior Geophysicist



Python Programming.



1 - Data Types Exploration

Escape Sequences / Characters

Python Escape Characters

Code	Result/Output	Description
\'	Single Quote	Add single quote with in a String
\\	Backslash	Insert single Back Slash
\n	New Line	\n takes the cursor to first position of a new line
\r	Carriage Return	\r takes the cursor to the first position of the same line
\t	Tab	\t add spaces of 8 characters
\b	Backspace	\b takes the cursor one position backward
\f	Form Feed	Form Feed is page breaking ASCII control character
\ooo	Octal value	Octal value
\xhh	Hex value	Hex value

❑ Assignment:

Generate this using one single line code:

- Basin Name is “GOS”
- Well Count is 956
- Main Reservoir is 'Miocene'

1 - Data Types Exploration

Strings

String Multiplication

❑ Assignment:

Generate this by creating 3 different variables:

output: 198 Wells are producing in GOS

1 - Data Types Exploration

Strings

Input Function

❑ Assignment:

Generate this by using one variable, input function and string formatting:

output: Loading Belayim Land data!

1 - Data Types Exploration

Strings

Input Function

❑ Assignment:

1. Make Computer ask you a question of 'how many fields you have in your company? and what are their names?'

2. Consider you have three fields. store the answer in appropriate variables.

3. print a full meaningful statement with your response using string formatting

1 - Data Types Exploration

Strings

String Slicing

❑ Fun Exercise:

1. The computer asks about your first name in reverse order using input function.

2. print ' Your name is : Your name in the right order '

❑ Assignment:

1. Make computer ask you about water saturation in well-x using input function.

2. print ' The Hydrocarbon saturation in well-x is value of Hydrocarbon Saturation '